

# Deterministic Signal-Processing Methods for Extracting Human Voice Features from Audio

## Executive summary

A strong **non-machine-learning** voice-analysis system should not be built around a single feature extractor. The most reliable design is a **multi-resolution, source-aware signal-processing pipeline** in which different estimators answer different physical questions: periodicity estimators recover vocal-fold vibration rate; LPC-family methods estimate vocal-tract resonances; Fourier/filter-bank representations describe spectral shape; cycle-synchronous measurements characterize perturbation; and inverse filtering attempts to separate glottal excitation from the vocal tract. These quantities are related, but they are not interchangeable. The classical source-filter model and short-time speech-processing framework remain the appropriate organizing abstractions. <sup>1</sup>

The central practical recommendation is to **retain at least two signal branches**:

1. a minimally processed branch for measurements whose physical meaning depends on the waveform itself—especially intensity, jitter, shimmer, pulse timing, and glottal-flow analysis; and
2. a conservatively enhanced branch for pitch, voicing, formants, endpoint/boundary detection, and many spectral descriptors.

This avoids a common methodological error: applying spectral subtraction, Wiener filtering, automatic gain control, or nonlinear noise suppression and then interpreting the resulting cycle-to-cycle amplitude or timing changes as properties of the speaker. Jitter is explicitly defined from differences between consecutive glottal periods, while shimmer is derived from consecutive-cycle amplitudes; therefore transformations that alter timing or amplitude can directly bias them. Praat likewise recommends voice-perturbation measures primarily on sustained phonation rather than arbitrary connected speech. <sup>2</sup>

For **F0/pitch**, the best deterministic general-purpose architecture is not “choose YIN or autocorrelation,” but **generate several physically different candidates and reconcile them**. Filtered autocorrelation or YIN should normally be the primary time-domain estimator; SWIPE supplies a complementary frequency-domain harmonic-template estimate; a cepstral estimate is useful as an inexpensive additional check. Autocorrelation has a long history in speech pitch estimation, YIN was specifically designed to reduce several common pitch-estimation errors, and SWIPE determines pitch by comparing spectra with a sawtooth-inspired harmonic pattern. Praat currently identifies filtered autocorrelation as its method of choice for intonation and vocal-fold vibration frequency. <sup>3</sup>

For **formants**, use pre-emphasized, short-window LPC/Burg analysis to generate resonance candidates and then impose **temporal continuity** using deterministic dynamic programming or state-space/Kalman filtering. Do not treat each frame independently. Praat's Burg implementation, for example, explicitly resamples relative to the formant ceiling, pre-emphasizes, windows the frame, estimates Burg LPC coefficients, converts poles to formants, and discards likely edge artifacts. Its documentation stresses that

the ceiling is speaker-dependent; typical examples are about 5 kHz for an adult male, 5.5 kHz for an adult female, and substantially higher for children. <sup>4</sup> Kalman-family trackers add continuity and uncertainty modeling; the KARMA work demonstrated the usefulness of this state-space perspective and reported lower overall formant-tracking error than conventional Praat/WaveSurfer trajectories on its evaluations. <sup>5</sup>

For **spectral characterization**, STFT, CQT and MFCCs are completely compatible with a no-ML requirement when used as deterministic signal descriptors. An STFT is simply a sequence of windowed FFTs and gives a linear-frequency local spectrum. <sup>6</sup> A CQT uses frequency bins whose center-frequency-to-bandwidth relationship is approximately constant, making it especially natural for harmonic/pitch structure. <sup>7</sup> MFCCs deterministically compress a log mel-filterbank spectrum through a DCT; they need not feed a classifier and can instead be interpreted as low-dimensional spectral-envelope coefficients.

For **prosodic and temporal analysis**, combine short-time energy/intensity, normalized periodicity, zero-crossing/spectral measures, pause detection, duration and rule-based syllable-nucleus detection. Classical speech processing has long used short-time energy and zero crossings for endpoint/speech-state decisions. <sup>8</sup> Exact phoneme identification from acoustics without a learned acoustic model is substantially harder. A defensible non-ML alternative is therefore to identify **acoustic landmarks and broad segment classes**, using abrupt spectral/energy changes, voicing transitions, bursts, frication onset/offset and formant continuity. Stevens' landmark framework provides the linguistic basis for this style of analysis. <sup>9</sup>

The principal workflow recommendations are:

| Condition           | Recommended core                                                                                                                                                                                                                                                                                                   |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Clean studio</b> | Minimal high-pass/DC removal → VAD → filtered ACF/YIN + SWIPE cross-check → Burg LPC + continuity/Kalman tracking → STFT/MFCC/CQT descriptors → cycle-synchronous jitter/shimmer → IAIF on selected voiced intervals                                                                                               |
| <b>Telephone</b>    | Preserve original channel bandwidth → pitch from ACF/YIN with SWIPE/cepstrum validation → primarily F1/F2 LPC tracking → STFT/MFCC → conservative temporal/prosodic measures; treat high formants, shimmer and glottal-flow parameters cautiously                                                                  |
| <b>Noisy field</b>  | Raw + enhanced parallel branches → adaptive noise estimate → conservative Wiener/spectral subtraction only on enhanced branch → multi-estimator pitch consensus → confidence-gated formants → median/Kalman trajectory smoothing → suppress unreliable jitter/shimmer/glottal estimates rather than forcing values |

These recommendations follow from the very different information losses in the three recording conditions. TIMIT provides broadband microphone speech and verified time-aligned phonetic labels; NTIMIT retransmits the same 6,300 recordings through real telephone handsets/networks; and Noisy TIMIT supplies the same material with several additive-noise families over 5–50 dB SNR, making this family particularly useful for controlled deterministic-method comparisons. <sup>10</sup>

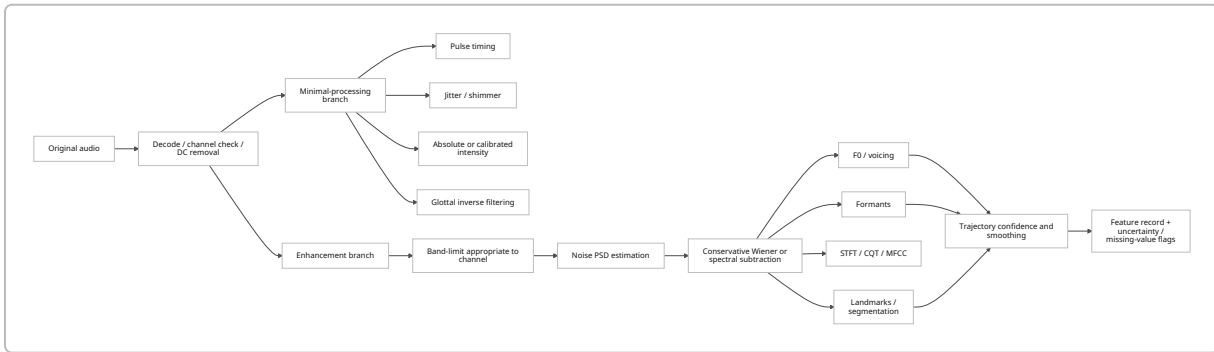
## Signal model, preprocessing, and measurement discipline

A useful abstraction for voiced speech is

$$s[n] \approx g[n] * v[n] * r[n] + \eta[n],$$

where  $g[n]$  represents glottal excitation,  $v[n]$  the vocal-tract response,  $r[n]$  lip radiation and recording/channel effects, and  $\eta[n]$  additive interference. IAIF is explicitly based on estimating and successively removing glottal, vocal-tract and radiation contributions; Alku's original description first models the average glottal spectral effect with a low-order all-pole filter, then obtains an LPC vocal-tract estimate, and finally inverse-filters the vocal tract and lip radiation to recover a glottal excitation estimate. <sup>11</sup>

This model immediately implies that preprocessing should be **feature-specific**. Removing a 60-Hz electrical contaminant may help an F0 estimator, but a high-pass corner placed above a low-pitched speaker's F0 may remove physically relevant source energy. A spectral enhancer may improve VAD while corrupting shimmer. A channel equalizer may improve spectral-envelope comparability while simultaneously changing absolute intensity. "One enhanced WAV for every downstream measurement" is therefore not a scientifically clean architecture.



## Preprocessing choices

**DC removal and high-pass filtering.** Remove DC first. For normal adult speech, a gentle high-pass substantially below the expected F0 range can remove handling/air-conditioning rumble without destroying periodicity. A practical starting range is roughly 30–70 Hz for ordinary adult speech, but this must be lowered for bass voices or other signals whose fundamental approaches the cutoff. This is an engineering range rather than a universal standard. Butterworth IIR filters are convenient because their order and critical frequencies are explicitly controllable; zero-phase forward/backward filtering is appropriate offline but is noncausal and therefore unsuitable when latency matters. [SciPy Butterworth documentation](#).

**Band-pass filtering.** Prefer band-limiting based on the **measurement** rather than generically filtering to "speech bandwidth." F0 estimation may benefit from suppressing irrelevant high-frequency noise, whereas formant analysis needs bandwidth through the highest formant of interest. Praat's filtered-autocorrelation pitch method deliberately low-pass filters before autocorrelation, and its documented filter progressively suppresses energy as it approaches the configured pitch top. <sup>12</sup>

**Telephone audio.** Never infer absent high-frequency vocal-tract information simply because the decoded file has a 16-kHz sample rate. NTIMIT, for example, is stored at 16 kHz but consists of original TIMIT utterances sent through telephone handsets and telephone-network channels; the physical channel, not the container sampling rate, determines available information. <sup>13</sup> For conventional narrowband telephone speech, upper formants can therefore be weak or unavailable even when the WAV Nyquist frequency is much higher.

**Spectral subtraction.** Classical spectral subtraction estimates a noise spectrum during nonspeech and subtracts an appropriately scaled version from the noisy short-time spectrum; Boll's 1979 method is the canonical reference. <sup>14</sup> It is inexpensive and useful with relatively stationary additive noise, but aggressive subtraction causes isolated residual spectral peaks—"musical noise"—and can alter spectral fine structure. For measurement, use the smallest subtraction strength that materially improves downstream confidence.

**Wiener filtering.** A deterministic short-time Wiener-style gain can be written approximately as

$$G(k, m) = \frac{\widehat{\text{SNR}}(k, m)}{1 + \widehat{\text{SNR}}(k, m)}, \quad \hat{S}(k, m) = G(k, m)Y(k, m),$$

with a recursively estimated noise PSD. The important engineering question is not whether the method is called Wiener or MMSE enhancement, but whether the gain is sufficiently smooth and conservative that formants, harmonics and amplitude modulation are not erased. Classical short-time spectral-amplitude estimators such as Ephraim–Malah provide a closely related foundational treatment of spectral-domain speech enhancement. <sup>15</sup>

**Median filtering.** Median filtering is especially useful **after feature extraction**: a 3–7-frame median on F0 candidates, formant tracks, energy derivatives or boundary scores can remove isolated impulses without smearing the entire waveform. Waveform median filtering should be used sparingly because it is nonlinear and can change zero crossings, pulse shape and high-frequency spectral content.

**Pre-emphasis.** It is particularly important for LPC/formants. Praat's Burg formant algorithm uses a +6 dB/octave-style pre-emphasis above a configurable frequency to compensate for the downward spectral tilt of vowel spectra so LPC poles better follow local resonance peaks rather than global source tilt. <sup>16</sup> This same preprocessing is **not** automatically appropriate for shimmer, calibrated intensity or glottal spectral-tilt measurements.

## Recommended signal branches

| Branch                 | Processing                               | Use                                           | Do not use blindly for                 |
|------------------------|------------------------------------------|-----------------------------------------------|----------------------------------------|
| Raw/<br>calibrated     | Decode, channel<br>selection, DC removal | SPL/intensity when<br>calibrated, waveform QA | Low-SNR pitch/formants                 |
| Minimally<br>filtered  | Gentle HP/BP only                        | Pulse marks, jitter,<br>shimmer, glottal flow | Harsh-noise<br>segmentation            |
| Enhanced               | Noise PSD + mild<br>Wiener/subtraction   | VAD, F0 candidates,<br>formants, landmarks    | Cycle-amplitude<br>perturbation        |
| Spectral-<br>analysis  | Windowing + optional<br>pre-emphasis     | STFT, LPC/Burg, MFCC                          | Absolute waveform<br>amplitude         |
| Telephone-<br>specific | Native channel retained                  | F0, F1/F2, timing                             | Claims about missing<br>upper spectrum |

The safest general principle is **do not denoise unless the expected reduction in estimator variance exceeds the bias introduced by the denoiser**. For physical voice features, returning “unreliable” is preferable to manufacturing a smooth numerical trajectory.

## Deterministic feature-extraction methods

### Pitch and pitch contour

For pitch analysis, the relevant acoustic quantity is usually the fundamental frequency  $F_0$ , even though psychoacoustic “pitch” and physical  $F_0$  are not strictly identical. The important implementation challenges are voicing decisions, harmonic/subharmonic ambiguity, octave errors, rapid transitions and short intervals.

**Autocorrelation.** For frame  $x[n]$ ,

$$R[\tau] = \sum_n x[n]x[n + \tau],$$

and the strongest plausible positive-lag peak gives

$$\hat{F}_0 = \frac{f_s}{\hat{\tau}}.$$

Normalized autocorrelation removes part of the amplitude dependence. Rabiner's classic work established autocorrelation as a practical speech-pitch method, while modern Praat uses a candidate-based autocorrelation tracker rather than simply taking the largest framewise peak. <sup>17</sup>

The key parameters are  $F_{0,\min}$ ,  $F_{0,\max}$ , frame length, hop, voicing threshold and continuity penalty. The lower pitch limit is particularly important because several periods must fit in the analysis window. Praat's filtered-autocorrelation implementation uses an effective window of three periods at the pitch floor; a 50-Hz floor therefore implies about 60 ms of effective analysis support. Its default zero time step corresponds to four estimates per window length. <sup>12</sup>

Autocorrelation is low-to-moderate cost: direct computation is  $O(NL)$  for  $L$  candidate lags, or about  $O(N \log N)$  using FFT correlation. Its principal errors are octave/subharmonic selections caused by non-sinusoidal excitation, strong spectral coloration and noise. Temporal candidate tracking is therefore substantially preferable to independent peak picking.

**YIN.** YIN replaces direct correlation peak selection with a squared difference function,

$$d(\tau) = \sum_j [x_j - x_{j+\tau}]^2,$$

followed by a cumulative-mean normalization that suppresses spurious short-lag minima. A threshold selects the first sufficiently strong period candidate, usually followed by local interpolation. The original JASA paper describes YIN as a fundamental-frequency estimator based on autocorrelation-related processing and was explicitly designed to improve several common failure modes of classical pitch estimators. <sup>18</sup>

Recommended starting values are 2–4 periods per analysis region, a 5–10 ms output hop, a speaker-appropriate F0 search interval, and a YIN threshold around 0.1–0.2 followed by validation rather than blindly accepting every minimum. Lower thresholds demand cleaner periodicity. These are practical ranges, not universal constants.

YIN tends to be strong on clean speech and singing and usually handles waveform shape better than simple peak-picked autocorrelation, but it still fails on breathy/irregular phonation, rapidly changing F0, strong additive noise and ambiguous subharmonics. Direct difference-function evaluation is roughly  $O(NL)$ ; FFT-accelerated variants reduce this.

**Cepstral pitch.** Define a real cepstrum approximately as

$$c[q] = \mathcal{F}^{-1} \{ \log(|X[k]| + \epsilon) \}.$$

A harmonic spectrum has approximately periodic spectral ripples, which become a peak near the pitch period in the quefrency domain. Search only

$$q \in \left[ \frac{1}{F_{0,\max}}, \frac{1}{F_{0,\min}} \right].$$

Noll's cepstral pitch work is one of the foundational references for this approach. 19

Its computational cost is low, roughly one FFT and inverse FFT per frame. It is a valuable independent checker because its failure mechanism differs from time-domain autocorrelation. Its weaknesses are spectral-envelope leakage, finite harmonic count, reverberation and ambiguity when a strong formant or channel response creates cepstral structure near the F0 quefrency.

**SWIPE.** SWIPE—Sawtooth Waveform Inspired Pitch Estimator—scores candidate F0 values by comparing the signal spectrum with a sawtooth-inspired harmonic kernel. The original Camacho–Harris work defines the estimate in terms of how well the observed spectrum matches the harmonic structure expected for a candidate fundamental. 20

Its advantage is integration of evidence across harmonics, making it a particularly valuable complement to time-domain methods. Its cost is higher than a single autocorrelation or cepstrum because multiple spectral resolutions/candidate frequencies may be evaluated. It is strong where harmonic structure is evident, but a strongly bandwidth-limited channel, severe spectral noise or too few resolved harmonics removes much of the evidence it exploits.

**Recommended deterministic F0 fusion.**

$$\text{accept } F_0 \iff \begin{cases} \text{voicing confidence high,} \\ \text{at least two estimators agree within tolerance,} \\ \text{transition from previous } F_0 \text{ is physically plausible.} \end{cases}$$

A useful log-frequency agreement measure is

$$e_{\text{cents}} = 1200 \left| \log_2 \frac{F_a}{F_b} \right|.$$

Agreement within roughly 25–50 cents is a conservative high-confidence rule; wider disagreement can be resolved using normalized autocorrelation/YIN periodicity, SWIPE strength and trajectory continuity. Do **not** simply average 100 and 200 Hz: that turns an octave disagreement into a physically unsupported 150-Hz estimate.

## Formants and vocal-tract resonances

**LPC.** Linear prediction models a sample by

$$x[n] \approx - \sum_{k=1}^p a_k x[n-k] + e[n],$$

giving an all-pole vocal-tract model

$$H(z) = \frac{G}{1 + \sum_{k=1}^p a_k z^{-k}}.$$

Complex-conjugate poles  $z_i = r_i e^{j\theta_i}$  yield candidate formants and bandwidths,

$$F_i = \frac{\theta_i f_s}{2\pi}, \quad B_i = -\frac{f_s}{\pi} \ln r_i.$$

Atal and Hanauer's foundational linear-prediction work formalized speech analysis/synthesis through this predictor/source representation. <sup>21</sup>

LPC works best when the speech segment is approximately stationary, voiced/sonorant, and sufficiently broadband. It can invent poles to explain spectral slope, split a broad resonance, merge nearby resonances, and confuse harmonics with resonances when F0 is high. This is why formant estimation should return confidence and missing values rather than always returning “F1–F5.”

**Burg LPC.** Burg estimation recursively minimizes forward and backward prediction errors without explicitly forming the conventional autocorrelation system. In speech tools it is attractive for stable, short-time all-pole estimation. Praat labels its Burg formant path as the preferred formant-analysis route and uses twice the requested maximum number of formants as the number of LPC poles. <sup>22</sup>

Praat gives unusually concrete parameter guidance: a typical human-speech setting is five formants per frame; its examples use a 5.5-kHz ceiling for an average adult woman, 5 kHz for an average adult man and up to about 8 kHz for a young child. A 25-ms effective analysis window has about 52-Hz —3 dB spectral resolution in its documented Gaussian-like window. It also stresses substantial interspeaker and intervowel variability, so these values must be tuned rather than treated as biological constants. <sup>16</sup>

For a generic implementation at 16 kHz, a useful starting regime is 20–30 ms windows, 5–10 ms hops, LPC order around 10–16, and a formant ceiling chosen from speaker/channel evidence. At 8-kHz telephone bandwidth, the model should be restricted to the observable range rather than using higher LPC order to invent unavailable resonances.

**Kalman tracking/smoothing.** Let the latent state contain formant frequency and optionally velocity,

$$\mathbf{x}_t = [F_{1,t}, F_{2,t}, F_{3,t}, \dot{F}_{1,t}, \dot{F}_{2,t}, \dot{F}_{3,t}]^\top.$$

Then

$$\mathbf{x}_t = A\mathbf{x}_{t-1} + \mathbf{w}_t, \quad \mathbf{y}_t = h(\mathbf{x}_t) + \mathbf{v}_t,$$

where observations  $\mathbf{y}_t$  may be LPC/Burg candidates. A forward Kalman filter imposes causal continuity; an offline Rauch–Tung–Striebel smoother uses future frames as well. In KARMA, the state-space formulation was taken further with an extended Kalman framework for jointly tracking formant/antiformant structure and associated uncertainty. <sup>5</sup>

The key tuning variable is the process/measurement-noise ratio. Too little process noise produces unrealistically smooth tracks and misses genuine transitions; too much collapses toward framewise LPC. A sensible deterministic adaptation is to increase observation variance when LPC poles have broad bandwidth, poor spectral-peak support or low frame SNR.

A Kalman smoother **cannot restore an unobserved formant**. Its role is candidate association, continuity enforcement and uncertainty propagation—not bandwidth extrapolation.

## Spectral descriptors

**STFT.** An STFT computes FFTs of successive windowed segments. SciPy's current `ShortTimeFFT` documentation defines the time step from the hop size and the bin width from FFT length and sampling period. <sup>6</sup>

For 16-kHz speech, 20–30 ms windows and 5–10 ms hops are good general starting points. A 25-ms frame contains 400 samples; using a 512- or 1024-point FFT changes spectral sampling density but does **not** create true resolution beyond that provided by the analysis window.

Useful deterministic STFT-derived features include spectral centroid, spread, roll-off, flux, spectral slope/tilt, harmonic-band energies, spectral flatness, band-energy ratios and peak trajectories. These should be retained with explicit units and normalization metadata.

**CQT.** The constant-Q transform places analysis bands on an approximately logarithmic frequency axis; Brown's original formulation is the foundational source. <sup>7</sup> Because human F0 variation and harmonic spacing are naturally multiplicative, CQT is useful for harmonic structure, pitch-range visualization and octave-normalized spectral summaries.

Typical analysis choices are 12 bins/octave for semitone-scale analysis, 24–36 when finer harmonic/pitch detail is desired, and a low-frequency limit chosen below the expected F0. The disadvantage follows directly from constant-Q geometry: low-frequency filters become temporally long and high-frequency bins relatively broad. CQT is therefore usually less convenient than linear-frequency STFT/LPC when exact upper-formant frequencies in hertz are the target.

**MFCCs as descriptors.** MFCC computation is deterministic:

$$|X[k]|^2 \rightarrow \text{mel filterbank} \rightarrow \log \rightarrow \text{DCT}.$$

Nothing about MFCCs intrinsically requires machine learning. The first coefficients summarize broad spectral-envelope shape;  $c_0$  or log-energy can represent overall level depending on convention. Retaining about 12–20 cepstral coefficients from roughly 20–40 mel bands is a conventional speech-analysis scale, but the exact number should reflect whether fine spectral detail or compact envelope characterization matters.

Their limitation is also their purpose: MFCCs discard local spectral detail. They should not substitute for explicit F0, formant bandwidth, H1–H2 or jitter measurements. Delta-MFCCs can be computed by deterministic finite differences and interpreted as envelope dynamics, but there is little benefit in calling them “features for recognition” when the actual goal is physical voice measurement.

## Energy, intensity, voicing and voice perturbation

### Energy/RMS.

$$E_t = \sum_n w[n] x_t[n]^2, \quad \text{RMS}_t = \sqrt{\frac{\sum_n w[n] x_t[n]^2}{\sum_n w[n]}}.$$

For relative prosody, use log-RMS or dB relative to a defined reference. For true SPL claims, microphone calibration is required. Praat's intensity analysis squares the signal and smooths it with a pitch-floor-dependent window; its method is therefore not identical to an arbitrary 25-ms RMS sequence. <sup>23</sup>

**Voiced/unvoiced detection.** A robust non-ML decision should combine evidence rather than use an energy threshold alone. Useful deterministic inputs are normalized ACF peak/YIN periodicity, harmonic-to-noise-like periodicity, energy above the local noise floor, ZCR, spectral flatness, and optionally an LPC spectral-distance measure. Classical voiced/unvoiced/silence work combines spectral/LPC and energy information, while endpoint algorithms historically use energy and zero-crossing information. <sup>24</sup>

For example,

$$S_v = w_1 \rho_{\text{ACF}} + w_2 (1 - \text{flatness}) + w_3 \text{SNR}_{\text{local}} - w_4 \text{ZCR}_{\text{norm}},$$

followed by two thresholds  $T_{\text{on}} > T_{\text{off}}$  gives hysteresis. The weights can be fixed from physical reasoning or calibration experiments; adapting thresholds from an estimated noise floor is still rule-based, not ML.

**Jitter.** Praat's local jitter is

$$J_{\text{local}} = \frac{\frac{1}{N-1} \sum_{i=2}^N |T_i - T_{i-1}|}{\frac{1}{N} \sum_{i=1}^N T_i}.$$

Praat allows period-floor, period-ceiling and maximum-period-factor rules so clearly implausible cycles do not dominate the statistic. Its documentation gives 0.1 ms as an example small period floor and notes that a 50-Hz minimum periodicity corresponds to a 20-ms period ceiling. <sup>25</sup> RAP, PPQ5 and DDP provide related perturbation definitions. <sup>26</sup>

**Shimmer** is analogous in amplitude rather than period, e.g.

$$S_{\text{local}} = \frac{\frac{1}{N-1} \sum_{i=2}^N |A_i - A_{i-1}|}{\frac{1}{N} \sum_i A_i}.$$

Praat provides local, dB and amplitude-perturbation-quotient variants. Both jitter and shimmer are normally measured on sustained vowels because cycle identification is substantially more trustworthy than in strongly articulating connected speech. <sup>27</sup>

Consequently, **jitter and shimmer should usually be omitted rather than “repaired” in low-SNR or codec-damaged speech**. Telephone AGC, packet/codec artifacts, clipping, reverberation and strong denoising can all change cycle amplitude or apparent pulse timing independently of physiology.

## Duration, speech rate and non-ML linguistic cues

Duration is straightforward only after segmentation. Recommended primitive events are speech onset/offset, pause onset/offset, voiced-interval onset/offset, vowel-like nucleus peaks, burst candidates and frication intervals.

A useful hierarchy is:

waveform → speech/non-speech → voiced/unvoiced → acoustic landmarks → broad segment classes.

**Speech duration** is total speech-active time. **Speaking rate** can be words or syllables divided by elapsed time including pauses. **Articulation rate** excludes designated pauses. Without a transcript or learned phone recognizer, syllable counts should be described as *estimated nuclei*, not ground-truth syllables.

A deterministic syllable-nucleus detector can smooth intensity/sonority, find local maxima, reject peaks whose surrounding spectral evidence looks unvoiced, impose a minimum peak-prominence criterion relative to the local noise floor, and merge peaks separated by less than roughly 80–150 ms. The exact refractory period must be validated for the language and speaking style.

For **phoneme-boundary candidates**, compute several framewise change functions:

$$D(t) = \alpha |\Delta \log E_t| + \beta \text{spectralFlux}_t + \gamma |\Delta F_{1,t}| + \delta |\Delta F_{2,t}| + \epsilon I[\text{voicing change}] + \zeta \text{burstScore}_t.$$

Candidate boundaries are local maxima of  $D(t)$ , followed by nonmaximum suppression and minimum-duration rules. Stevens' landmark theory explicitly motivates the use of abrupt acoustically salient events for lexical/phonetic analysis rather than assuming that every phoneme boundary is equally evident in the signal. <sup>9</sup>

The method is particularly reasonable for:

- stop release bursts and closure transitions;
- frication onset/offset;
- strong voiced/unvoiced transitions;
- silence boundaries;
- vowel/sonorant transitions accompanied by major formant or energy change.

It is intrinsically weaker for smoothly coarticulated boundaries, adjacent vowels/approximants and other transitions with no abrupt landmark. This is a limitation of acoustic observability, not merely an algorithmic deficiency.

**Strict no-ML alternatives to conventional forced alignment.** Modern “forced alignment” generally means transcript-constrained inference using an acoustic model. Under the present constraint, three legitimate alternatives remain: manual/semiautomatic alignment with rule-generated boundary candidates; deterministic DTW against manually selected speaker- or phoneme-specific reference templates; and pronunciation/duration rules coupled to acoustic-landmark constraints. These can force a known transcript into temporal order, but they should not be expected to match the generality of a trained acoustic model. TIMIT’s hand-verified, time-aligned phoneme and word labels make it an appropriate reference for measuring how much is lost under such purely deterministic alignment. <sup>28</sup>

## Source–filter separation and glottal flow

Classical inverse filtering attempts

$$\hat{G}(z) = S(z) \hat{V}^{-1}(z) \hat{R}^{-1}(z).$$

The difficulty is that both  $G(z)$  and  $V(z)$  are unknown; estimating the vocal tract incorrectly immediately contaminates the glottal estimate.

**IAIF** resolves this iteratively. Alku’s original algorithm first obtains a low-order estimate of the glottal spectral contribution, removes it, estimates the vocal tract using LPC, then inverse-filters vocal tract and radiation effects. PSIAIF makes the process pitch-synchronous. <sup>11</sup>

Useful outputs include estimated glottal-flow waveform, glottal-flow derivative, closure timing, open/closed-phase proxies, normalized amplitude quotient, spectral tilt and harmonic-amplitude measures. These are only as trustworthy as the source/filter separation.

Practical orders commonly use a **low-order glottal model** and a higher-order vocal-tract LPC model. The COVAREP `iaif_gci` implementation exposes vocal-tract and glottal LPC orders, GCI-synchronous analysis and leaky integration; its implementation uses values such as a 0.99 leaky-integration factor as a typical setting. <sup>29</sup> GFM-IAIF modifies IAIF’s glottal spectral model to reduce bias in the source estimate while retaining the iterative source/filter structure. <sup>30</sup>

Inverse filtering is best restricted to **high-SNR voiced intervals**. High F0, low F1, vocal-tract/source coupling, nasalization, reverberation and inaccurate GCIs all undermine the assumptions. In field audio it is often scientifically preferable to report only F0/periodicity and spectral-tilt proxies rather than claim a precise glottal waveform.



## Comparative engineering matrix and recommended workflows

The ratings below are **engineering synthesis rather than universal benchmark scores**. “Accuracy” means ability to recover the intended physical quantity under favorable conditions; it should not be interpreted as a dataset-independent numerical error rate. Complexity is per analysis stream, assuming ordinary speech frame sizes. The literature itself shows strong dependence on speaker, SNR, F0 range and parameterization, so fixed “95% accurate” claims would be misleading. <sup>31</sup>

### Pitch and resonance methods

| Method          | Principle                                 | Clean accuracy | Noise robustness | Latency                                 | Approx. cost                   | Useful parameter range                                               | Dominant limitations                                              |
|-----------------|-------------------------------------------|----------------|------------------|-----------------------------------------|--------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|
| Autocorrelation | Periodic lag similarity                   | High           | Medium           | ~2–3+ F0 periods                        | $O(NL)$ , FFT<br>$O(N \log N)$ | F0 bounds; 2–4+ periods; 5–15 ms hop                                 | Octave/subharmonic errors; noise/reverb <sup>17</sup>             |
| Filtered ACF    | LP filtering + ACF candidates             | High           | Med-High         | $\geq 3$ periods in Praat default logic | Low-med                        | Praat default floor 50 Hz, top 800 Hz; effective window $3/F_{\min}$ | Top/floor must fit speaker; still periodicity-based <sup>12</sup> |
| YIN             | Difference function + CMND + threshold    | High           | Medium           | ~2–4 periods                            | $O(NL)$ direct                 | threshold $\approx 0.1$ – $0.2$ starting range; explicit F0 bounds   | Irregular phonation, SNR, fast transitions <sup>32</sup>          |
| Cepstrum        | Peak in quefrency domain                  | Med-High       | Medium           | one analysis frame                      | $O(N \log N)$                  | search $q = 1/F_0$ ; ~20–50 ms frames                                | Formant/channel cepstral leakage; few harmonics <sup>33</sup>     |
| SWIPE           | Harmonic/sawtooth spectral template match | High           | Med-High         | multi-resolution                        | Medium-high                    | dense log-F0 candidates; suitable harmonic bandwidth                 | Bandwidth loss; spectral noise; heavier compute <sup>20</sup>     |

| Method              | Principle                        | Clean accuracy          | Noise robustness | Latency   | Approx. cost        | Useful parameter range                                | Dominant limitations                                         |
|---------------------|----------------------------------|-------------------------|------------------|-----------|---------------------|-------------------------------------------------------|--------------------------------------------------------------|
| LPC formants        | All-pole vocal-tract model       | High on suitable vowels | Low-med          | ~20–30 ms | $O(Np + p^2)$       | order ~10–16 at 16 kHz; ceiling speaker-specific      | High F0, noise, nasal zeros, order sensitivity <sup>34</sup> |
| Burg formants       | Forward/backward LPC estimation  | High                    | Medium           | ~20–30 ms | $O(Np)$ -scale      | Praat: ~5 formants, ceiling ~5–5.5 kHz adult examples | Same all-pole assumptions; false/split poles <sup>16</sup>   |
| Kalman filter       | Stateful temporal continuity     | Improves tracks         | Med-High         | causal    | $O(d^3)$ , tiny $d$ | process/obs variance from candidate confidence        | Oversmoothing if $Q$ too small                               |
| RTS/Kalman smoother | Forward/backward state smoothing | High offline            | Med-High         | noncausal | $O(Td^3)$           | same + full/partial future context                    | Cannot restore missing formants <sup>5</sup>                 |

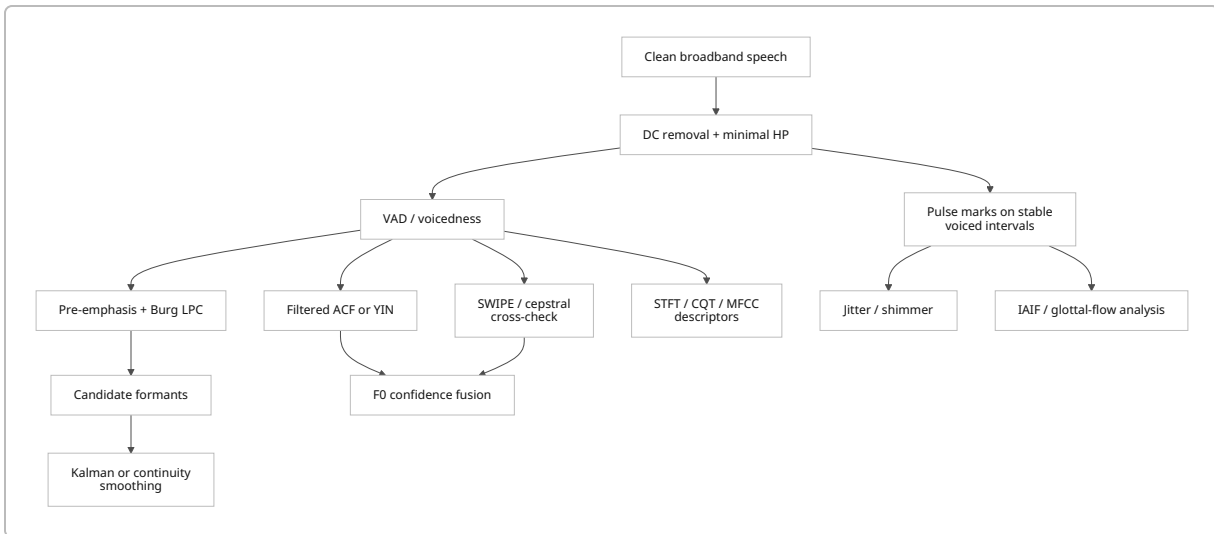
## Spectral, prosodic and source methods

| Method | What it measures                 | Noise robustness | Latency                 | Complexity | Typical configuration                            | Important caveat                                                          |
|--------|----------------------------------|------------------|-------------------------|------------|--------------------------------------------------|---------------------------------------------------------------------------|
| STFT   | Local linear-frequency spectrum  | Medium           | window length           | Low        | 20–30 ms, 5–10 ms hop, Hann/Gaussian-like window | FFT zero-padding increases sampling, not physical resolution <sup>6</sup> |
| CQT    | Log-frequency spectral structure | Medium           | high at low frequencies | Med-high   | 12–36 bins/octave                                | Less natural for exact high-frequency formant Hz estimates <sup>7</sup>   |

| Method                | What it measures                         | Noise robustness | Latency                 | Complexity                | Typical configuration                         | Important caveat                                         |
|-----------------------|------------------------------------------|------------------|-------------------------|---------------------------|-----------------------------------------------|----------------------------------------------------------|
| MFCC                  | Compact log spectral envelope            | Med-high         | STFT-scale              | Low                       | ~20–40 mel bands, ~12–20 coefficients         | Deliberately loses narrow spectral detail                |
| RMS/log energy        | Amplitude/prosodic envelope              | Medium           | low                     | $O(N)$                    | 10–30 ms analysis                             | Gain/AGC changes interpretation                          |
| Intensity             | Smoothed power in dB                     | Medium           | pitch-floor dependent   | Low                       | choose pitch floor below speaker F0           | Absolute SPL requires calibration <sup>23</sup>          |
| Rule-based V/UV       | Periodicity + energy + spectral evidence | Med-high         | low                     | Low                       | hysteretic adaptive thresholds                | Fixed global thresholds generalize poorly <sup>35</sup>  |
| Local jitter          | Cycle-period perturbation                | <b>Low</b>       | several cycles          | Low after pulse detection | sustained periodic vowel                      | Extremely pulse-error sensitive <sup>36</sup>            |
| Local shimmer         | Cycle-amplitude perturbation             | <b>Low</b>       | several cycles          | Low                       | sustained periodic vowel                      | Gain/noise/denoising sensitive <sup>37</sup>             |
| IAIF/PSIAIF           | Glottal source via inverse filtering     | Low-med          | frame/pitch synchronous | Medium                    | low-order source + higher-order tract model   | Source/filter errors are coupled <sup>11</sup>           |
| Landmark segmentation | Abrupt acoustic transitions              | Medium           | ~1–several frames       | Low-med                   | energy derivative + flux + voicing + formants | Smooth coarticulation has no sharp boundary <sup>9</sup> |

## Clean studio workflow

For high-SNR studio speech, enhancement usually creates more risk than benefit. Use the original PCM recording, remove DC, apply only a gentle feature-specific high-pass, and keep levels unchanged for perturbation features. TIMIT's 16-kHz, 16-bit microphone recordings provide a representative controlled broadband test environment. <sup>28</sup>



Recommended priorities are:

**F0:** filtered ACF or YIN primary; SWIPE independent check; interpolate only brief high-confidence gaps.

**Formants:** 20–30 ms Burg windows, speaker-appropriate ceiling, temporal state tracking and bandwidth/peak-support validation. Praat's documented 5-kHz/5.5-kHz adult ceiling examples are reasonable starting points, not mandatory values. <sup>16</sup>

**Voice quality:** use stable vowel nuclei or deliberately recorded sustained vowels. Do not mix connected-speech jitter/shimmer with sustained-vowel reference ranges. <sup>38</sup>

**Source analysis:** IAIF/GFM-IAIF only on voiced high-SNR intervals with reliable pulse timing. <sup>39</sup>

## Telephone workflow

NTIMIT is especially useful here because it contains the *same underlying TIMIT utterances* passed through actual telephone handsets/network paths and time-aligned back to the original recordings. It thus permits paired broadband-versus-telephone evaluation without changing lexical content or speakers. <sup>13</sup>

The recommended pipeline is:

native telephone waveform → channel-aware VAD → {F0, F1/F2, spectral envelope, timing}.

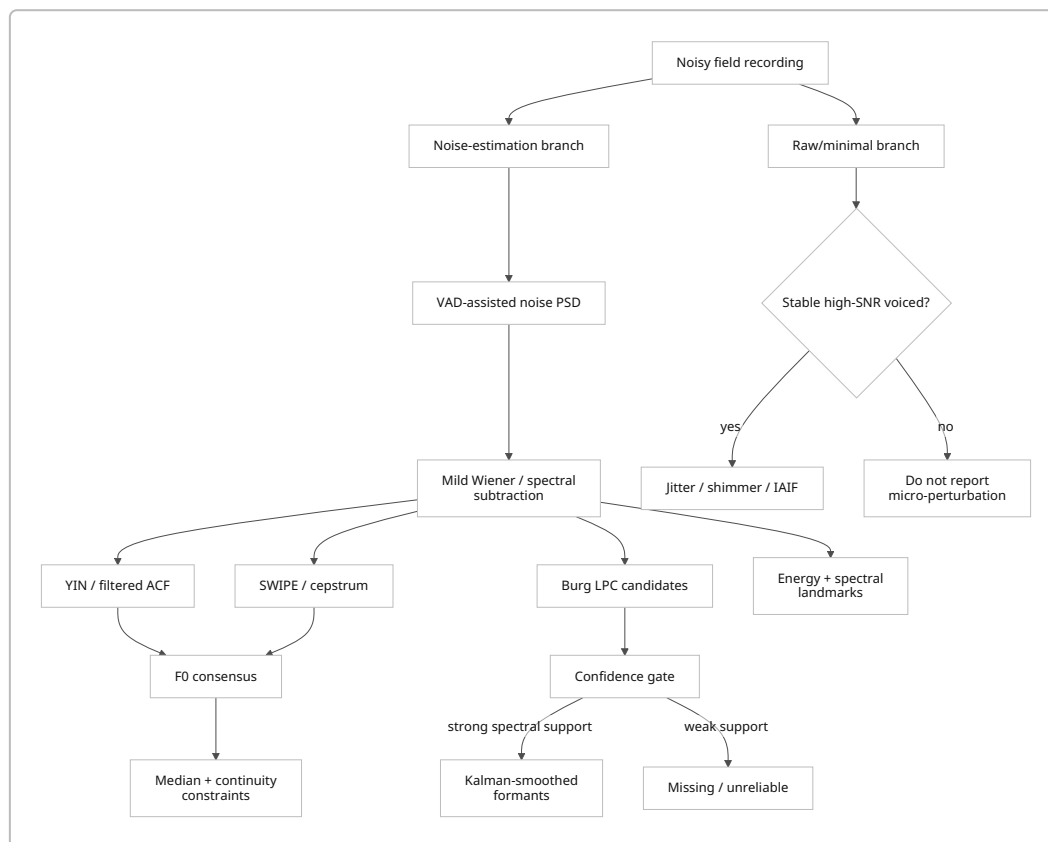
Do not upsample and then interpret newly interpolated frequencies as recovered speech bandwidth.

Pitch remains relatively feasible because ordinary speech F0 lies well inside telephone passbands, although codecs, clipping and channel ripple can increase periodicity errors. Use ACF/YIN with trajectory constraints, and let spectral/cepstral estimators act as validators rather than assuming all estimators remain equally informative.

For formants, emphasize F1 and F2. Report F3 only when actual spectral support exists; do not report higher formants merely because LPC returns poles. For voice-quality work, regard shimmer and glottal spectral measures as channel-dependent unless the entire acquisition path is standardized.

## Noisy-field workflow

The key architecture is **confidence gating plus parallel branches**, not aggressive restoration. Noisy TIMIT gives controlled white, pink, blue, red, violet and babble additions over 5–50 dB SNR and is therefore useful for constructing estimator-failure curves. 40



For impulsive contamination, first remove obvious clipped/burst intervals from estimation rather than attempting to reconstruct them. For stationary noise, conservative Wiener/spectral subtraction is appropriate. For nonstationary babble, the benefit of enhancement is smaller and multi-estimator agreement is particularly important.

Median filtering should generally be applied to **candidate trajectories** rather than the waveform. Formant and F0 smoothing should never bridge long unvoiced intervals. Every final frame should carry at least: estimate, voiced/speech state, method agreement and quality/confidence flag.

## Evaluation metrics and test datasets

No deterministic method should be evaluated solely by how “smooth” its outputs look. Smoothness can conceal octave errors, incorrect formant identity and overaggressive interpolation. Evaluation should separately score **local estimation**, **detection/classification**, **temporal tracking**, and **failure calibration**.

### Pitch metrics

Use at least four quantities:

$$\text{relative error} = \frac{|\hat{F}_0 - F_0|}{F_0},$$

$$e_{\text{cent}} = 1200 \log_2(\hat{F}_0 / F_0),$$

plus a **gross pitch error** rate under a declared threshold, and **voicing decision error**.

A 20% relative threshold is common in speech-F0 evaluation literature for defining a gross error, but the threshold should always be stated explicitly rather than treated as an invariant property. Report octave-error rate separately because octave mistakes are qualitatively different from a small local bias.

For reference periods, simultaneous electroglottographic/laryngographic recording is substantially more useful than another acoustic pitch estimator. PTDB-TUG was created specifically as a pitch-tracking corpus with synchronized microphone and laryngograph signals; the associated corpus publication describes 2,342 phonetically rich utterances. <sup>41</sup> A laryngograph measures vocal-fold-contact-related conductance changes and can supply period timing independent of the microphone acoustic waveform. <sup>42</sup>

### Formant metrics

For each formant  $i$ ,

$$\text{MAE}_i = \frac{1}{N} \sum_t |\hat{F}_i(t) - F_i(t)|,$$

$$\text{RMSE}_i = \sqrt{\frac{1}{N} \sum_t (\hat{F}_i(t) - F_i(t))^2}.$$

Also report:

- identity swaps  $F_i \leftrightarrow F_{i+1}$ ;
- percentage within a declared tolerance in Hz or percent;
- missing-value rate;
- gross outlier rate;
- trajectory derivative error;
- results stratified by speaker F0, sex/age group, vowel and SNR.

Hillenbrand et al.'s American English vowel study is valuable for static/trajectory formant validation: it contains 45 men, 48 women and 46 children, with F1–F4 measured from LPC spectra using interactive

editing. The authors explicitly used human inspection rather than treating raw LPC poles as ground truth.

<sup>43</sup> That distinction is important: an automatically generated LPC track should not be evaluated against another automatically generated LPC track and then called physiologically validated.

## Segmentation, VAD and landmarks

For speech/non-speech or voiced/unvoiced decisions, report precision, recall, F1, confusion matrices and duration-weighted error. For boundaries, use

$$\Delta t_j = \hat{t}_j - t_j$$

and report median absolute boundary error plus F1 under several tolerances such as  $\pm 10$ ,  $\pm 20$  and  $\pm 30$  ms. Multi-tolerance reporting prevents a detector from looking good merely because the scoring window is generous.

TIMIT is particularly suitable because its orthographic, word and phonetic transcriptions are time-aligned and hand-verified. It contains about five hours from 630 speakers, ten phonetically rich sentences per speaker, at 16 kHz/16 bit. <sup>28</sup>

For deterministic acoustic-landmark research, additionally score boundary types separately: closure, release, frication, voicing onset/offset and sonorant/vowel transition. Theoretical landmark work predicts that not all segment boundaries are equally acoustically salient. <sup>9</sup>

## Jitter, shimmer and glottal metrics

For jitter/shimmer, a useful validation hierarchy is:

reference pulse timing  $\rightarrow$  period/amplitude sequence  $\rightarrow$  perturbation statistic.

Report pulse-detection error separately from perturbation-value error; otherwise a method can appear to have a “jitter problem” when the actual failure is period marking. Because jitter explicitly depends on adjacent cycle periods, even a single missed/doubled pulse can create a large local error. <sup>25</sup>

Use MAE/bias, within-speaker coefficient of variation, test-retest reliability and sensitivity curves versus SNR. Do not combine sustained vowels and connected speech into the same error distribution without stratification. Praat's voice-analysis guidance treats sustained vowels as the typical measurement context. <sup>38</sup>

For glottal inverse filtering, where synthetic or simultaneous reference information is available, measure waveform correlation or normalized RMSE, GCI timing error, spectral-distortion error, and error in derived source parameters. The OPENGLOT/GIF evaluation ecosystem and IAIF/GFM-IAIF implementations are useful references for controlled source-filter experiments. <sup>44</sup>

## Recommended datasets

| Dataset                       | Best use                                                         | Why it is useful                                                                                             | Limitation                                                              |
|-------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>TIMIT</b>                  | Boundaries, duration, clean phonetics, formant/pitch experiments | 630 speakers; 6,300 utterances; 16-kHz broadband; hand-verified phone/word timing <sup>28</sup>              | Read US English; not physiological ground truth                         |
| <b>NTIMIT</b>                 | Telephone/channel robustness                                     | Same TIMIT material retransmitted over real telephone handsets/networks and realigned to TIMIT <sup>13</sup> | Historical telephone network; not modern codec diversity                |
| <b>Noisy TIMIT</b>            | Controlled SNR/noise curves                                      | ~322 h; six additive noise types; 5–50 dB in 5-dB steps <sup>40</sup>                                        | Additive noise does not capture all field reverberation/channel effects |
| <b>FFMTIMIT</b>               | Far/free-field robustness                                        | TIMIT rerecorded through a free-field path; useful for room/channel effects <sup>45</sup>                    | Still derived from read speech                                          |
| <b>WTIMIT</b>                 | Wideband mobile-channel effects                                  | TIMIT transported through real mobile telephony conditions <sup>46</sup>                                     | Channel-specific                                                        |
| <b>PTDB-TUG</b>               | F0/voicing                                                       | Synchronized microphone and laryngograph; designed for pitch evaluation <sup>41</sup>                        | Primarily a pitch reference, not complete vocal-tract ground truth      |
| <b>Hillenbrand vowel data</b> | Formants/vowel trajectories                                      | Manually inspected F1–F4 on men, women and children <sup>43</sup>                                            | Isolated h-V-d vowels, not arbitrary connected speech                   |

A particularly strong experiment is a **paired-channel design**:

$$\text{TIMIT} \rightarrow \begin{cases} \text{NTIMIT} & \text{telephone} \\ \text{FFMTIMIT} & \text{free field} \\ \text{Noisy TIMIT} & \text{controlled additive noise} \end{cases}$$

because much of the linguistic and speaker variability is inherited from the same base corpus. The LDC documentation explicitly identifies these derived TIMIT collections and their channel/noise manipulations.

<sup>47</sup>

## Experimental protocol

For rigorous comparison, freeze parameters **before** the final evaluation set. Parameter adaptation may be deterministic and recording-specific—noise-floor estimation, pitch-range narrowing, speaker-specific formant ceiling—but the adaptation rule itself must be fixed.

Report results by SNR, F0 quartile, speaker, voiced-segment duration, telephone/broadband condition and speech class. This exposes the characteristic failure regimes: high-F0 formant errors, low-SNR voicing failures and short-segment pitch instability.

Most importantly, evaluate **abstention**. For physical voice measurement, an estimator that returns valid values on 85% of frames with a 1% gross-error rate can be more scientifically useful than one that fills 100% of frames by extrapolation but has many silent gross failures. Confidence calibration should therefore be treated as a first-class metric.

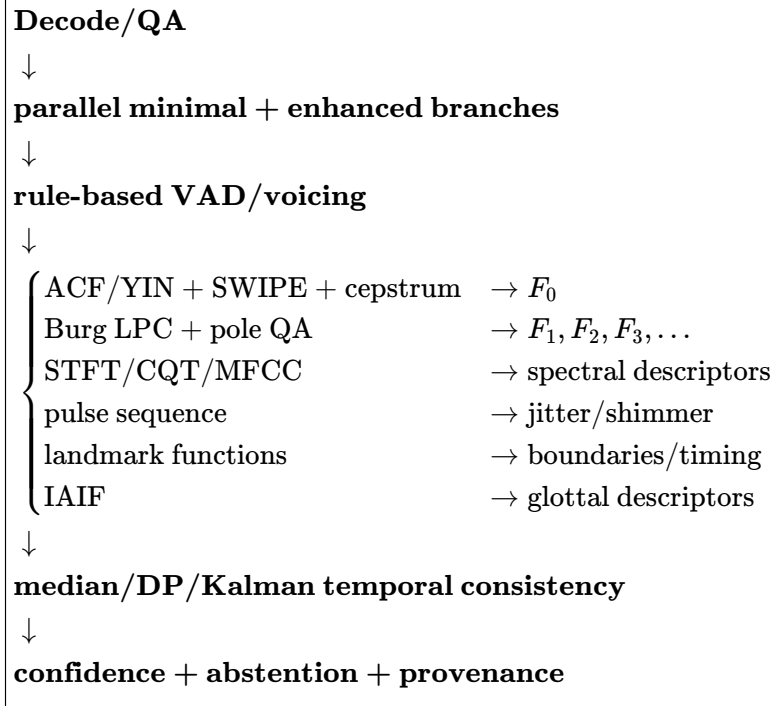
## Reference implementations, primary sources, and recommended architecture

The following resources are appropriate starting points because they are either primary publications or authoritative implementation documentation. They are not ML models.

| Area                           | Primary / authoritative reference                                      | Implementation or documentation                                                          |
|--------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Filtered autocorrelation pitch | Boersma/Praat methodology <sup>12</sup>                                | <a href="#">Praat filtered autocorrelation</a>                                           |
| Raw autocorrelation pitch      | Rabiner's autocorrelation work <sup>48</sup>                           | <a href="#">Praat</a>                                                                    |
| YIN                            | de Cheveigné & Kawahara, <i>JASA</i> , 2002 <sup>32</sup>              | <a href="#">librosa 0.11 yin</a> ; <a href="#">aubio pitch API</a>                       |
| SWIPE                          | Camacho & Harris, <i>JASA</i> , 2008 <sup>20</sup>                     | Use the paper as normative algorithm reference; independently validate third-party ports |
| Cepstral F0                    | Noll, "Cepstrum Pitch Determination" <sup>33</sup>                     | FFT/IFFT implementation directly in SciPy/NumPy                                          |
| Burg formants                  | Praat documented Burg pipeline <sup>16</sup>                           | <a href="#">Praat To Formant (burg)</a>                                                  |
| LPC formants                   | Atal & Hanauer linear prediction <sup>21</sup>                         | <a href="#">MathWorks LPC formant example</a>                                            |
| Kalman formants                | KARMA, Mehta et al. <sup>49</sup>                                      | Paper/state-space equations are preferable to treating a tracker as a black box          |
| STFT                           | Standard time-frequency analysis; SciPy authoritative API <sup>6</sup> | <a href="#">SciPy ShortTimeFFT</a>                                                       |
| CQT                            | Brown, 1991 <sup>7</sup>                                               | <a href="#">librosa 0.11 CQT</a>                                                         |

| Area                 | Primary / authoritative reference                | Implementation or documentation                                                                    |
|----------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------|
| MFCC                 | Classical mel-filterbank/cepstral representation | <a href="#">librosa 0.11 MFCC</a>                                                                  |
| Filtering            | Standard Butterworth design                      | <a href="#">SciPy</a> <code>butter</code> , <code>zero-phase</code> <a href="#">sosfiltfilt</a>    |
| Spectral subtraction | Boll, 1979 <sup>14</sup>                         | Straightforward STFT-domain implementation; preserve an unenhanced branch                          |
| Intensity            | Praat algorithm <sup>23</sup>                    | <a href="#">Praat intensity</a>                                                                    |
| Jitter               | Praat definitions <sup>36</sup>                  | <a href="#">Praat local jitter</a>                                                                 |
| Shimmer              | Praat definitions <sup>37</sup>                  | <a href="#">Praat Voice analysis</a>                                                               |
| IAIF/PSIAIF          | Alku, 1991 <sup>11</sup>                         | <a href="#">COVAREP</a> , including <code>iaif_gci.m</code> <sup>29</sup>                          |
| Acoustic landmarks   | Stevens, 2002 <sup>9</sup>                       | Implement as explicit energy/spectral/voicing/formant change rules                                 |
| Endpoint detection   | Rabiner & Sambur, 1975 <sup>50</sup>             | Energy/ZCR/periodicity implementation is simple and auditable                                      |
| Evaluation speech    | LDC TIMIT family                                 | <a href="#">TIMIT catalog</a> , <a href="#">NTIMIT</a> , <a href="#">Noisy TIMIT</a> <sup>10</sup> |
| Pitch ground truth   | PTDB-TUG                                         | <a href="#">TU Graz SPSC datasets/tools</a> <sup>51</sup>                                          |

The architecture I would choose for a research-quality implementation is therefore:



The final feature record should preserve not merely numerical values but the **measurement conditions**: sampling rate, channel type, preprocessing, F0 bounds, analysis window/hop, LPC order and formant ceiling, denoising parameters, voicing confidence, estimator agreement, and whether a value was observed, smoothed, interpolated or rejected. This matters particularly for formants, where Praat explicitly documents strong dependence on formant ceiling and vocal-tract differences, and for pitch, where the pitch floor directly determines analysis-window length. <sup>52</sup>

For a single recommended non-ML stack, the highest-value combination is **filtered autocorrelation/YIN + SWIPE consensus for F0; Burg-LPC candidates + Kalman/continuity tracking for formants; STFT plus MFCC for general spectrum and CQT when harmonic detail matters; energy + periodicity + spectral rules for VAD; pulse-synchronous jitter/shimmer only in qualified voiced intervals; and IAIF only where SNR and periodicity make source-filter separation defensible**. That combination uses genuinely different physical views of the same voice rather than accumulating redundant variants of one estimator, and it provides natural cross-checks that a single classical algorithm cannot supply. <sup>53</sup>

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<sup>1</sup> <sup>11</sup> <sup>39</sup> ISCA Archive - Glottal wave analysis with pitch synchronous iterative adaptive inverse filtering  
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